Mirror twins facilitator guide

Mathematical overview

Object and motions. Use congruent rigid regular tetrahedra (the kit's card models) with letters at the corners. Allowed matching motions are translations and rotations in three-dimensional space; vertices and labels stay attached. Letter direction and decoration do not count. A reflected picture is used to construct a comparison model; reflection is not an allowed physical matching move.

Orientation obstruction. Four distinct vertex labels give a chiral object: its mirror image cannot match it by an allowed motion. Align A and look from A toward the center of the opposite face, with A nearest the eye on both models. The cyclic order of B, C, D is preserved under rotations and reversed by reflection. A failed search proves impossibility only when it covers every turn: A must go on A, and trying the three turns about A is complete.

Counts. A bare regular tetrahedron has exactly 12 proper rotational symmetries, counting the identity: 4 possible destinations for one vertex and 3 compatible placements of the others. Equivalently, there are the identity, eight 120°/240° vertex-axis rotations and three 180° opposite-edge-axis rotations. Distinct motion histories count as the same rotation when all vertex destinations agree. The 24 assignments of four distinct labels form two rotational classes of 12, exchanged by reflection.

Repeated labels. Making any two of the four labels equal makes the mirror pair match. More generally, all assignments of the multiset A, A, B, C are rotationally equivalent. An odd vertex permutation taking one labeling to another can be made even by swapping the indistinguishable vertices. Restoring distinct labels can produce either rotational class. This is a fact about this vertex-colored regular tetrahedron, not a claim that repeated colors remove chirality from every object.

Before classroom use

Mandatory gate, not yet performed: an adult must test the actual rigid models: align a true copy, construct and compare the intended four-label mirror pair, verify the specified viewing-direction certificate, and successfully align the two-equal-label example. Mathematical and computational checks do not replace this physical pretest. If the card models or the completed pretest cannot be supplied, defer classroom use; do not substitute a paper-only activity for K-1.

What children are meant to discover

- **K-1:** Match genuine rotated copies, notice the difficulty of the mirror pair and investigate what changes when labels repeat. No complete impossibility proof is expected.
- **Grades 2-3:** Distinguish failure to find a turn from a complete check of the three turns with A on A; compare fresh pairwise label merges and recover both outcomes when distinct labels return.
- **Grades 4-5:** Use the consistent cyclic-order certificate, explain all six merges, count the 12 vertex maps and the two labeling classes.

Status. Unpiloted. The card models and classroom timing remain untested.

Preparation and the physical gate

Materials: the printed kit

Print **week-37-materials.pdf** at 100% and check its 10 cm line. Counts are for eleven children (K-1 4, grades 2-3 4, grades 4-5 3) at three adult-led tables; the grades 4-5 trio compares two of its three models at a time.

Models, sheet 1, card stock, 13 sheets: a net with 90 mm edges, one model per child and two for the adults (pretest, launch, spare). Cut, score with a ballpoint against a ruler, fold with the printed side in, and tape the three open edges. A closed card tetrahedron is rigid; it is the “frame” the student pages mention. Leave the faces blank.

Letter caps, sheet 2, card stock, 4 sheets (one per table): each sheet gives four models A, A, B, B, C, C, D. A cap is three small triangles that wrap one corner, so its letter shows on all three faces; a pea-sized ball of sticky tack holds it, and changing a letter means swapping the cap. The spares let a pair make any two letters equal while keeping one name, which every merge in grades 2-3 and 4-5 Problem 4 needs (the B-D merge needs the second B), and build A, A, B, C.

Pose mat, sheet 3, copy paper, 7 sheets: two face-size footprints marked back, front left and front right. With both models standing on footprints, “same pose” is a corner-by-corner check, and the printed poses can be copied directly.

Also: scissors, a ruler, clear tape and sticky tack; a paper trimmer helps with the cap rows.

Preparation estimate: about 6 minutes per model, so about 1¼ hours for thirteen, plus 30-40 minutes to cut and fold the caps; children in grades 2-5 can fold their own caps during the launch. These are estimates: the kit is digitally checked but physically unrehearsed.

Required pretest on the actual models

1. True copies: label two models A at top, B front left, C front right and D back, matching the first diagram. Rotate one out of sight, then recover complete agreement without relabeling. Confirm that both models can be held in the same pose without flexing.

2. Mirror pair: keep the first model and place A top, C front left, B front right, D back on the second. Check the back vertex rather than trusting a flat silhouette. Try matching; use the orientation test next to verify that failure reflects the intended mathematical obstruction.

3. Viewing direction: on both models look from A toward the center of face BCD, A nearest the eye. Read B, C, D around the face in the same chosen direction. The orders must be opposite, up to a cyclic starting shift. Looking through the face toward A on one model invalidates the comparison.

4. Repeated labels: on this original mirror pair swap the D cap for a C cap on both. Rotate one about the A-to-opposite-face axis until B matches B. Both C vertices must then match C. Reset labels and check another merge if time permits.

Record the tester, date, model type and outcome before class. These steps are instructions for a future pretest; no one has yet reported performing it. If the supposed mirror pair matches with four distinct labels, or the repeated pair fails, stop and resolve construction, geometry, viewing direction or an extra unintended label mark before teaching.

Common launch and K-1 key

Flexible hour

0-8 minutes: allow handling, then stand two true copies on a pose mat and demonstrate one successful turn using the printed before/after figure. Say what counts as matching: every same-letter corner, regardless of the letter's direction. Demonstrate swapping a cap and keeping the caps on while turning. A successful copy match must precede the impossible comparison.

8-22 minutes: pairs hide and turn copies, then build the four-label mirror comparison with an adult checking the back vertex. Let children try several useful turns without prolonged frustration. **22-26:** movement break, using hands or bodies only to discuss “same view” and “other side,” not as a replacement tetrahedron model.

26-44 minutes: investigate repeated labels. Middle groups restore the original four-label pair before each merge. Upper groups can use the A-nearest-eye view. **44-56:** compare one successful match with one justified obstruction; upper children may count vertex destinations. **56-60:** restore and store models; note the tasks reached and any model or viewing confusion.

A successful physical match is already a complete existence demonstration. A failed search proves impossibility only when it covers every turn: A has to go on A, and then there are only three turns to try. Keep the cyclic-order argument optional until the question has become meaningful. An appropriate stopping point for younger children is discovering that the repeated-label pair can match.

K-1 answers

Problem 1: the two models are true copies and always match after undoing the hidden turn. If they do not, inspect the labeling and rigidity. Having the same top label is insufficient; all other corners must also agree.

Problem 2: the printed A B C D versus A C B D pair cannot match under rigid translations and rotations. It swaps B and C while keeping A and D in place in the construction. Accept a complete check: A must go on A, and all three turns about A fail. A young child is not expected to produce it; the adult can demonstrate the stable reversed order after exploration.

Problem 3: yes. After D becomes C, the pair is A B C C versus A C B C in the same positional order. A one-third turn around the A axis sends the first model's front-left B to the second's front-right B, while the two Cs occupy the two C positions. Choose the turning direction that aligns B; the other labels then agree.

Problem 4: no impossible pair exists with exactly A, A, B, C on congruent regular models. Every assignment belongs to one rotational class. Children can demonstrate examples; page 5 supplies the all-cases reason.

Problem 5: a matching pair can use identical labelings, with one model subsequently rotated. An unmatching pair is the four-label mirror pair from Problem 2. Check all corners, including the back.

Optional hints: for a true copy, “Put the A corners together in the same pose. Where is B?” For the mirror pair, “When A is nearest your eye on both, what order do you meet the other labels?” For repeated labels, “Which of these two identical corners would you like to match first?” Avoid claiming that turning a model upside down is forbidden; it is an ordinary allowed 3D rotation.

Keys for the older bands

Grades 2-3

Problem 1: true copies always match; undoing the hidden rigid motion is one matching motion.

Problem 2: the mirror pair cannot match. “No turn has worked yet” does not by itself establish “no turn can work.” Accept a complete check: A must go on A, and all three turns about A fail. The A-nearest-eye cyclic order is a second reason, covering every allowed turn.

Problem 3: answers depend on the pair built. Four distinct labels produce two classes: same-class pairs match, opposite-class pairs do not. The examples in K-1 Problem 5 supply both outcomes. Do not automatically call every newly made pair a mirror pair.

Problem 4: D becoming C allows a match, as above. Every other pairwise merge also allows a match: AB, AC, AD, BC, BD and CD. Start each test from the original A B C D / A C B D mirror pair; successive cumulative merges test a different question. The name kept for the merged pair does not change the result. Split the six merges between the pairs and pool results on the board.

Problem 5: all A A B C assignments match by rotation, so a partner cannot make a nonmatching pair under the stated model rules.

Problem 6: both outcomes are possible. In the printed pair, the first model has A at top and front left; the second has A at top and front right. Change the top A on both to D: the new pair does not match. Change the top A on the first and the front-right A on the second: it matches. Changing both side As gives a nonmatch; changing first-side and second-top gives a match. More generally, first align the repeated-label pair; restoring D at corresponding vertices gives a match, while restoring it at opposite choices of the repeated pair gives a nonmatch.

Grades 4-5

Problem 1: all four corners must agree. Once A is aligned, the cyclic placement of the other three labels matters, not merely which labels are present or which label is at the top.

Problem 2: no rigid matching motion exists for the four-distinct-label mirror pair. Page 5 gives both a geometric and an algebraic obstruction.

Problem 3: in the printed view from A, the original opposite face reads B, C, D counterclockwise, up to choice of starting place. The mirror reads B, D, C counterclockwise. A planar rotation of the view only changes the starting place, not this orientation. Use A nearest the eye on both; viewing one from the opposite side reverses the comparison and can create a false match.

Problem 4: all six unordered choices of two labels allow a rotational match after merging, with the original mirror pair restored each time. One proof uses the reflection swapping the two equal vertices, then composes it with the mirror comparison to obtain an orientation-preserving symmetry. Details follow on page 5.

Problem 5: 12 rotations as vertex maps, including leaving the model still. Only the final mapping counts, regardless of the motion history.

Problem 6: two genuinely different four-label models up to rotation. All 24 distinct assignments split into two classes of size 12. They are the two mirror forms; reflection is not allowed when deciding whether two assignments count as the same.

Proofs and source notes

The orientation certificate

Suppose a proper rigid motion matched the four-label models. It sends A to A and the center of the opposite face to the corresponding center. Looking along that same directed axis with A nearest the eye, the induced orientation on the opposite face is preserved. Its three labels can cycle under an axial turn but cannot reverse order. The two mirror models have reversed order. Thus no proper rigid motion can match them. This covers rotations about arbitrary axes.

An adult algebraic certificate is the nonzero determinant sign of vectors AB, AC, AD: translations and rotations with determinant +1 preserve it; reflection reverses it.

Why repetition permits a match

A regular tetrahedron has a reflection S exchanging any specified vertex pair and fixing the other two. Its plane contains the other vertices and the pair's midpoint. When the exchanged labels are equal, S preserves the labeled object. Compose S with the mirror isometry R: R composed with S is orientation-preserving and matches the mirror. Every pairwise merge works.

For any two assignments of A, A, B, C, choose a vertex bijection respecting labels. If it is odd, swap the two indistinguishable As in that bijection; it becomes even without altering the labeled target. Even vertex permutations are exactly the proper tetrahedral rotations. This proves that all such assignments match, while restoring distinct labels removes the freedom to make that parity adjustment.

Why the counts are complete

Choose where one starting vertex goes: four choices. Once it is aligned, the other three vertices lie at the corners of an equilateral opposite face; a proper rotation permits three cyclic placements, giving $4 \times 3 = 12$. Their explicit types are identity, two nonidentity turns about each of four vertex axes, and one half-turn about each of three opposite-edge axes: $1 + 8 + 3 = 12$. A four-distinct-label assignment has no nonidentity label-preserving rotation, so every orbit has 12 assignments. There are $4! = 24$ assignments in total, hence two orbits.

Sources and verification

Richard Schwartz, **Symmetry Groups of Platonic Solids** (Brown, 2007), §3, printed p. 3, **Rotations of the Tetrahedron**. Author-hosted handout. Source for the regular-coordinate model and rotation group A_4; the count here is independently checked.

OpenStax, **Organic Chemistry**, §5.1, Enantiomers and the Tetrahedral Carbon: tetrahedral mirror distinction. Givental, Nemirovskaya and Zakharevich, **Math Circle by the Bay** (2018), Preface pp. viii-ix: manipulatives and explanations; this guide's pacing is a local untested proposal.

A fresh determinant check of all 24 vertex permutations verifies 12 proper maps, two distinct-label orbits, all six merges and all four restore-D outcomes. It keys final student v2 without changes. **The actual-model pretest on page 2 remains unperformed.**

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

After the pretest on the card models succeeds, a first visit may end with matching true copies and the repeated-label pair (K-1 P1-3). On return, K-1 can make repeated-label and restored-label pairs (P4-5). Grades 2-3 can compare fresh label merges and then restore distinct labels (P4-6). Grades 4-5 can choose the viewing-direction certificate (P3), the all-merge explanation (P4), or the rotation/class counts (P5-6) as separate investigations. Keep the model specification and orientation check from guide pp1-2; the pretest remains unperformed here.